

EGB320 Final Report: Mobility Subsystem

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1 Problem Description

The problem set forward by the client was that shelves needed restocking, and it was believed that a robotic system would provide a more cost-effective method for shelf restocking. To evaluate the suitability of a robotic warehousing system a Technology Readiness Level 3(TRL3) prototype. This prototype was designed to work in a scaled down mock warehouse with shelves, items and obstacles that would simulate the major factors around navigating and moving stock around a grocery store.

For this prototype some adjustments were made to allow for more streamlined development. The first major adjustment was to have each different element in the store be represented by an object of a different color and shape as shown below:

Element	Representation
Obstacle	Green Colouring
Item	Orange Colouring
Aisle and Picking Station Marker	Black Colouring
Shelves	Blue Colouring
Picking Station	Yellow Colouring

These adjustments were made to allow for a more simplified vision system which in turn would allow for rapid development. Further prototypes will build on this model to allow for more complex vision processing. Along with this the shelf heights were the same for all aisles. Allowing for simplified item placement and navigation code as no matter the bay the heights will always remain the same. The final simplification is the distance between the picking stations, this distance allows for a lower level accuracy being required to pick up an item.

With these simplifications in mind 4 separate subsystems were identified that would be designed to integrate to a final product. The intention behind this was to further simplify the system through having each system capable of operating separately with methods of communication to allow for integration. This allowed for the 4 subsystems to be designed in parallel and setbacks in one element of the robot would not necessarily set back the overall progress of the robot.

Following this the requirements for each subsystem were identified, for mobility these requirements were as follows:

- Turn to a bearing on a single point in a given direction
- Turn at a set speed in a given direction
- Move forward and backward at a set speed
- Move forward and backward a set distance
- Have a chassis that is capable of carrying all other subsystems
- Have a power management system that can power all other subsystems

With these requirements identified background research was conducted to determine the most suitable solutions.

2 Background Research

Research was conducted into various methods for robotic mobility used in similar applications in industries. Four major methods of mobility were identified:

- Wheels
- Tracks
- Legs
- Mecanum Wheels

With these options identified investigation into the suitability of each method for the proposed application.